



2024 YEAR 12 TASK 1

Mathematics Extension 2

**General
Instructions**

- Reading time – 10 minutes
- Working time – 90 Minutes
- Write using blue or black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total Marks:
48**

Section I – 5 marks

- Attempt Questions 1–5
- Allow about 15 minutes for this section

Section II – 44 marks

- Attempt Questions 6–8
- Allow about 75 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Use the multiple-choice answer sheet for Questions 1–5.

- 1** Consider the vectors $\underline{a} = 3\underline{i} - 2\underline{j} + \underline{k}$ and $\underline{b} = -3\underline{i} + \underline{j} - 3\underline{k}$

Which of the following is closest to the obtuse angle between $\underline{a}$ and $\underline{b}$?

A. 0.5°

B. 2.6°

C. 1.3°

D. 1.8°

- 2 A parallelogram $OABC$ has the vertices at A and B defined by the position vectors

$$\begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}. \text{ Let } M \text{ be the midpoint of } \overrightarrow{BC}.$$

Which of the following is the vector $\overrightarrow{OM}$?

A. $\frac{1}{2} \begin{bmatrix} 8 \\ -1 \\ 7 \end{bmatrix}$

C. $\frac{1}{2} \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix}$

B. $-\frac{1}{2} \begin{bmatrix} 8 \\ -1 \\ 7 \end{bmatrix}$

D. $-\frac{1}{2} \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix}$

- 3 Gladys is looking at the relationship between two lines, $L_1 = a_1 + \lambda b_1$ and $L_2 = a_2 + \alpha b_2$ where λ and α are constants. She finds the following pieces of information.

- The ratio of a_1 to a_2 is 3.
- When solved simultaneously, there is no solutions.
- The dot product of b_1 and b_2 is not zero.

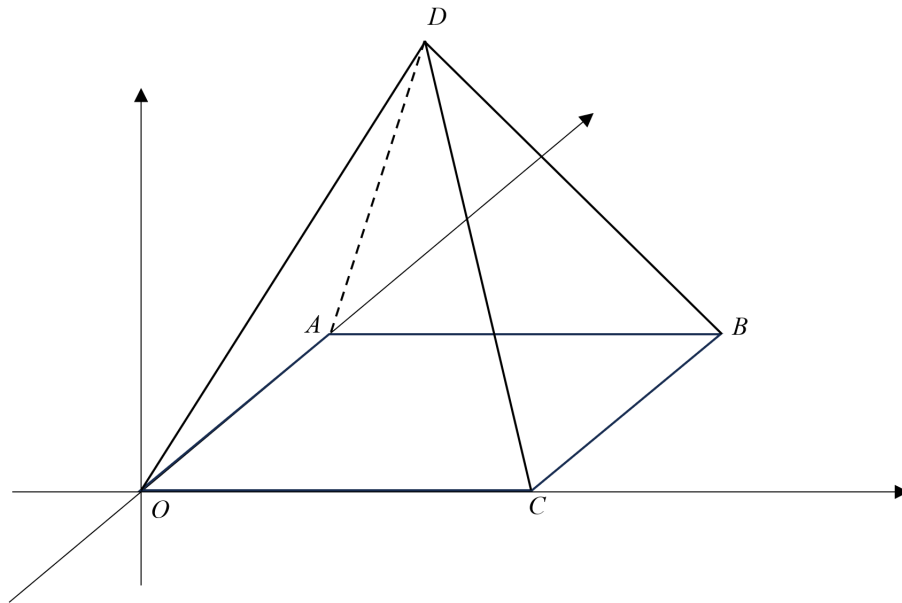
Based on her information only, which of the following relationships between L_1 and L_2 is guaranteed to be true?

- A. Parallel.
 - B. Perpendicular.
 - C. Skew.
 - D. None of the above.
- 4 Consider three unique unit vectors, a , b and c . It is known that $a \cdot b = 0$ and $c \cdot b = 0$.

Which of the following is false?

- A. $a \cdot c = 1$
- B. $(a - b) \cdot (c - b) = 0$
- C. $(a - c) \cdot b = 0$
- D. $|a - c| = 2 \times |b|$

- 5 A square based, right pyramid $OABCD$ is placed with its square face in the xy plane as shown below. All sides of the pyramid are unit length.



Let M be the point on the triangular face, CBD , that is equidistant from B , C and D .

What is the vector $\overrightarrow{OM}$?

- A. $\frac{1}{6} (5\mathbf{i} + 3\mathbf{j} - \sqrt{2}\mathbf{k})$
- B. $\frac{1}{6} (5\mathbf{i} - 3\mathbf{j} - \sqrt{2}\mathbf{k})$
- C. $\frac{1}{6} (5\mathbf{i} + 3\mathbf{j} + \sqrt{2}\mathbf{k})$
- D. $\frac{1}{6} (5\mathbf{i} - 3\mathbf{j} + \sqrt{2}\mathbf{k})$

End of Section I

Section II

43 marks

Attempt Questions 6 – 8

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 6 (14 marks) Please begin a new Writing Booklet.

- (a) Let $\underline{a} = 6\underline{i} - 3\underline{j} + \underline{k}$ and $\underline{b} = -4\underline{i} - 2\underline{j} - 3\underline{k}$.
- (i) Find $3\underline{a} - 2\underline{b}$. 1
- (ii) Evaluate $\underline{a} \cdot \hat{\underline{b}}$ showing full working. 2
- (iii) It is known that $\lambda\underline{a} + \mu\underline{b}$ is a vector parallel to the y/z plane. Find the smallest possible positive integer values of λ and μ . 1
- (b) The points $A = (3, 2, 7)$, $B = (1, -3, -3)$ and $C = (\lambda, 5, 13)$ are co-linear. Find the value of λ . 2
- (c) Consider the following vectors $\underline{a} = 2\underline{i} - \underline{j} + 4\underline{k}$ and $\underline{b} = -2\underline{i} - 2\underline{j} + 3\underline{k}$.
- (i) Find the vector projection of $\underline{b}$ onto $\underline{a}$. 2
- (ii) Hence or otherwise, find the values of λ and ζ where $\underline{b} = \lambda\underline{a} + \zeta$ and $\underline{a} \cdot \zeta = 0$ 2

(d) Consider the position vectors $\vec{OA} = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$, $\vec{OB} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$ and $\vec{OC} = \begin{bmatrix} 3 \\ 3 \\ -1 \end{bmatrix}$.

(i) Find the parametric equations of the line passing through A and B . **3**

(ii) Does C lie on the line found in part (i)? Justify your answer. **1**

End of Question 6

Question 7 (15 marks) Please begin a new Writing Booklet.

- (a) Consider the following two lines.

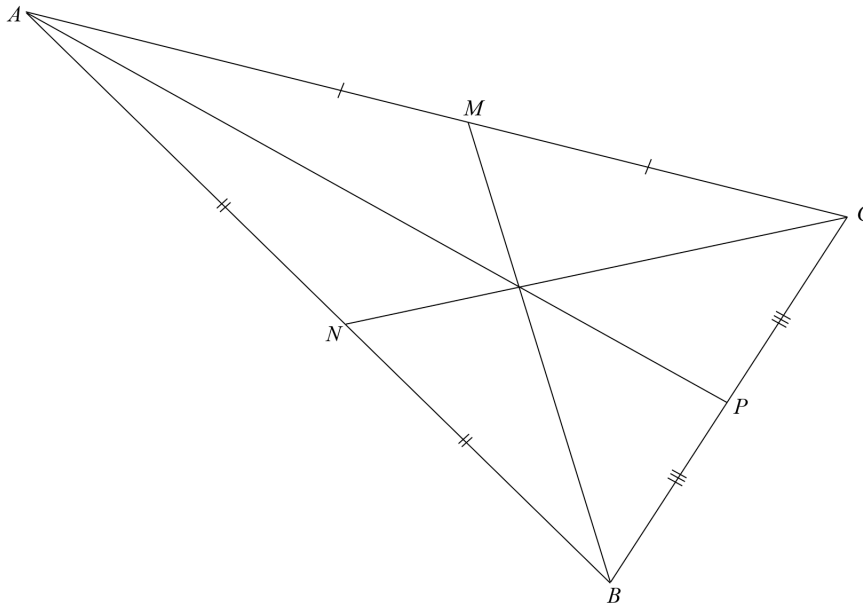
$$L_1 : \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \text{ and } L_2 : \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix} + \mu \begin{bmatrix} 3 \\ -4 \\ -2 \end{bmatrix}.$$

- (i) If it exists, find the point of intersection between L_1 and L_2 . **3**

- (ii) Hence or otherwise state the relationship/s between L_1 and L_2 justifying your answer. **1**

- (b) Consider $\triangle ABC$, where A , B and C have position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively.

The points M , N and P are the midpoints of AC , AB and BC respectively.



- (i) Show that $\overrightarrow{BM} = \frac{1}{2}(\underline{a} + \underline{c}) - \underline{b}$. **1**

- (ii) Show that $\overrightarrow{BM} + \overrightarrow{CN} + \overrightarrow{AP} = \mathbf{0}$ **2**

- (c) The plane P_1 contains the points $A = (1, 4, -2)$, $B = (-3, 0, 6)$ and $C = (-1, 2, -4)$
- (i) Find the Cartesian equation of the P_1 . **3**
- (ii) P_2 is a different plane that contains the points R , S and T . However, those three points are not sufficient to find the equation of P_2 . State the relationship between R , S and T . **1**
- (iii) P_3 is another plane with Cartesian equation $3x - 2y + z = 5$. Find the acute angle between P_1 and P_3 correct to the nearest degree. **2**
- (d) State the centre and radius of the sphere $2x^2 + 2y^2 + 2z^2 - 4x + y - 12z = -2$. **2**

End of Question 7

Question 8 (14 marks) Please begin a new Writing Booklet.

(a) Consider the vector equation $\underline{r}(t) = \begin{bmatrix} 2 \sin t \\ 2 \cos t \\ \sin 2t \end{bmatrix}$.

(i) Find the Cartesian equation of the curve. **1**

(ii) Find the cartesian equation of the tangent to the curve when $t = \frac{\pi}{6}$ **3**

(iii) Find the point/s at which the tangent to the curve is parallel to the x-y plane. **1**

(b) Consider the following two lines.

$$\underline{r}_1 = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \text{ and } \underline{r}_2 = \begin{bmatrix} 4 \\ 0 \\ -2 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} \text{ where } \lambda, \mu \in \mathbb{R}.$$

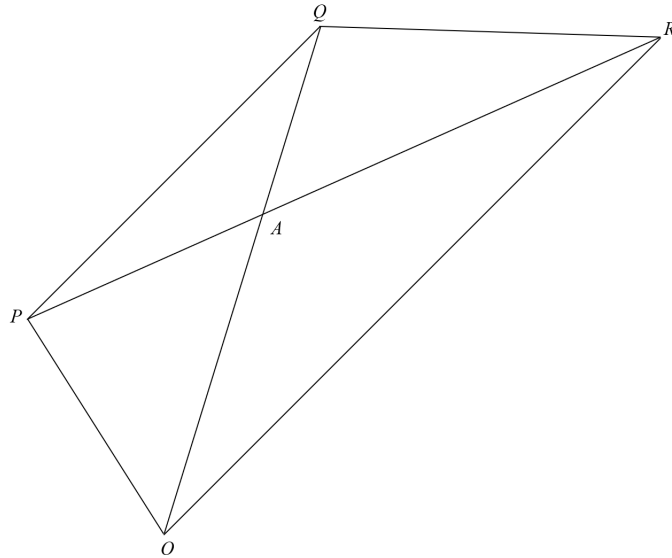
Let the point P lay on $\underline{r}_1$ with parameter p and let the point Q lie on $\underline{r}_2$ with parameter q .

(i) Find $\overrightarrow{PQ}$ in terms of p and q . **1**

(ii) Find the values of p and q when $\overrightarrow{PQ}$ is perpendicular to both $\underline{r}_1$ and $\underline{r}_2$. **3**

(iii) Hence find the minimum distance between the lines. **1**

- (c) In the diagram, $OPQR$ is a trapezium where $\vec{OP} = \underline{p}$, $\vec{OR} = \underline{r}$ and $\vec{OR}$ is parallel to $\vec{PQ}$.
Let the diagonals of the trapezium intersect at A .



- (i) Show that the vector equation of the line OQ can be written as $\underline{l}(t) = t(\underline{p} + \lambda \underline{r})$ **1**
where $t, \lambda \in \mathbb{R}$.
- (ii) Prove by vector methods that the point of intersection of the diagonals of a **3**
trapezium lies on the line passing through the midpoint of the parallel sides.

End of Exam

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

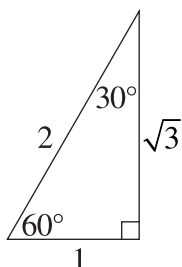
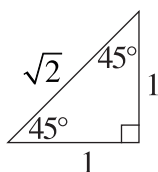
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

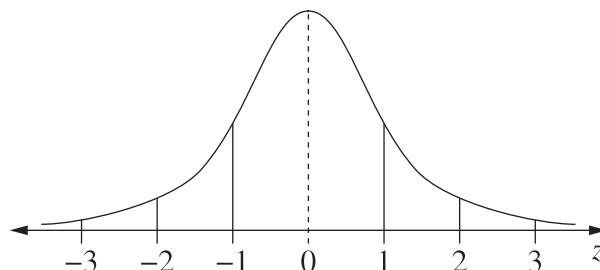
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

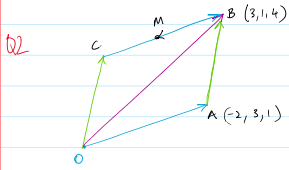
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Q1 $|a| = \sqrt{14}$ $|b| = \sqrt{19}$

$$\begin{aligned} \underline{a} \cdot \underline{b} &= -9 - 2 - 3 \\ &= -14 \end{aligned}$$

$$\cos \theta = \frac{-14}{\sqrt{14} \times \sqrt{19}}$$

$\theta \div 2.6029 \dots \therefore \textcircled{B}$



$$\vec{OM} = \vec{OB} + \vec{BM} \quad \vec{BM} = -\frac{1}{2}(\vec{OA})$$

$$= \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 4 \\ -\frac{1}{2} \\ \frac{7}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 8 \\ -1 \\ 7 \end{bmatrix} \therefore \textcircled{A}$$

Q3 (D) $\underline{b_1} \cdot \underline{b_2} \neq 0 \therefore$ Not perpendicular

No solution $\hat{=}$ No point of intersection

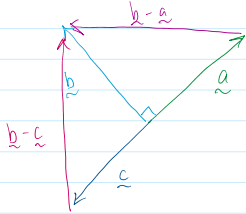
$a_2 : a_1 = 3:1$ Not direction vector
∴ no relationship.

∴ Can't tell if Skew or Perpendicular

∴ Not enough information. (D)

Q4 $\underline{a} \cdot \underline{b} = 0$ $\underline{b} \cdot \underline{c} = 0$ unit vectors. $\underline{a} \neq \underline{c}$

$\underline{a}$ & $\underline{c}$ must be parallel but in opposite directions.



$\therefore a \cdot c = -1$ $\therefore$ A is not true.

$$(b - c) \cdot (b - a) = 0 \quad \text{Nat } (b)$$

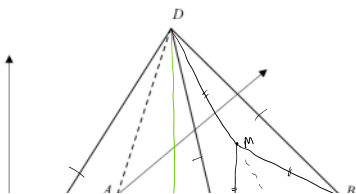
$$(a - c) = 2a$$

$$2\vec{a} \cdot \vec{b} = 0 \quad (\text{As they are perpendicular}) \therefore \text{Not } (C)$$

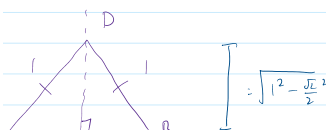
$$|a - c| = 2$$

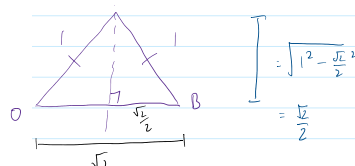
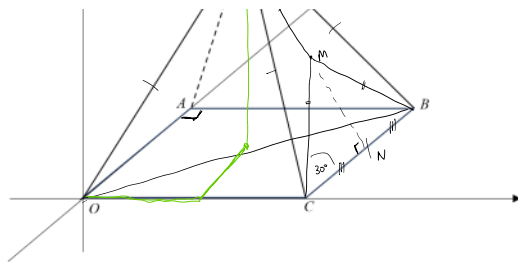
$$|b| = 1 \quad \therefore |a - c| = 2|b| \quad \therefore \text{Not (A)}$$

QS



$$|OB| = \sqrt{2}$$





$$\therefore \vec{OO} = \frac{1}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{1}{\sqrt{2}}\underline{k}$$

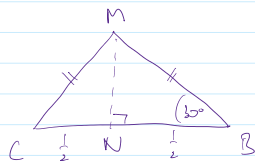
Let N be the midpoint of $\vec{CB}$

$$\vec{CN} = \frac{1}{2}\underline{j}$$

$$\vec{ND} = -\frac{1}{2}\underline{i} + \frac{1}{\sqrt{2}}\underline{k}$$

$$\vec{NM} = \lambda \vec{ND}$$

In $\triangle CMB$



$$\frac{|\vec{NM}|}{\frac{1}{2}} = \tan 30^\circ$$

$$|\vec{NM}| = \frac{1}{2} \times \frac{1}{\sqrt{3}} = \frac{1}{2\sqrt{3}}$$

$$\therefore |\vec{NM}| = \frac{1}{2\sqrt{3}}$$

$$|\vec{ND}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \frac{\sqrt{3}}{2}$$

$$\therefore \vec{NM} = \left(-\frac{1}{2}\underline{i} + \frac{1}{\sqrt{2}}\underline{k}\right) \div \frac{\sqrt{3}}{2} \times \frac{1}{2\sqrt{3}} = \frac{1}{3}\left(-\frac{1}{2}\underline{i} + \frac{1}{\sqrt{2}}\underline{k}\right)$$

$$\begin{aligned} \therefore \vec{OM} &= \vec{OC} + \vec{CN} + \vec{NM} \\ &= \underline{i} + \frac{1}{2}\underline{j} + \left(-\frac{1}{6}\underline{i} + \frac{1}{3\sqrt{2}}\underline{k}\right) \\ &= \frac{5}{6}\underline{i} + \frac{1}{2}\underline{j} + \frac{1}{3\sqrt{2}}\underline{k} \\ &= \frac{1}{6}\left[5\underline{i} + 3\underline{j} + \sqrt{2}\underline{k}\right] \end{aligned}$$

$$(a) \quad \underline{a} = 6\underline{i} - 3\underline{j} + \underline{k} \quad \underline{b} = -4\underline{i} - 2\underline{j} - 3\underline{k}$$

$$(i) \quad 3\underline{a} - 2\underline{b} = \begin{bmatrix} 3 \times 6 - 2 \times -4 \\ 3 \times -3 - 2 \times -2 \\ 3 \times 1 - 2 \times -3 \end{bmatrix}$$

$$= \begin{bmatrix} 26 \\ -5 \\ 9 \end{bmatrix}$$

$$(ii) \quad \underline{a} \cdot \underline{b} = (6 \times -4 + 3 \times 2 - 1 \times 3) \div \sqrt{16 + 4 + 9}$$

$$= \frac{-21}{\sqrt{29}} \quad (1)$$

$$(iii) \quad \lambda \underline{a} + \mu \underline{b} = \begin{bmatrix} 6\lambda - 4\mu \\ -3\lambda - 2\mu \\ \lambda - 3\mu \end{bmatrix}$$

Parallel to the y/z plane
 $\therefore x$ component is zero.

$$6\lambda - 4\mu = 0$$

$$\lambda = \frac{2}{3}\mu$$

Smallest positive solution $\mu = 3, \lambda = 2$ (1)

$$(b) \quad \overrightarrow{AB} = \begin{bmatrix} 1-3 \\ -3-2 \\ -3-1 \end{bmatrix} = \begin{bmatrix} -2 \\ -5 \\ -10 \end{bmatrix}$$

$$\overrightarrow{AC} = \begin{bmatrix} \lambda-3 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 6 \end{bmatrix}$$

$$\overrightarrow{BC} = \begin{bmatrix} \lambda-1 \\ 8 \\ 16 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ 16 \end{bmatrix}$$

$$\text{let } \overrightarrow{AC} = k \overrightarrow{AB}$$

$$\text{let } \overrightarrow{BC} = m \overrightarrow{AB}$$

$$\therefore \begin{bmatrix} \lambda-3 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -2k \\ -5k \\ -10k \end{bmatrix}$$

$$\therefore \begin{bmatrix} \lambda-1 \\ 8 \\ 16 \end{bmatrix} = \begin{bmatrix} -2k \\ -5k \\ -10k \end{bmatrix}$$

$$(\text{from } j) \quad k = -\frac{3}{5}$$

$$(\text{from } j) \quad k = -\frac{8}{5}$$

Only one needed.

$$\therefore \lambda - 3 = -2 \times \frac{-3}{5} \quad \therefore \lambda = \frac{21}{5} \quad (1)$$

$$\therefore \lambda - 1 = +16/5 \quad \lambda = \frac{21}{5}$$

$$(c) \quad \underline{a} = 2\underline{i} - \underline{j} + 4\underline{k} \quad \underline{b} = -2\underline{i} - 2\underline{j} + 3\underline{k}$$

$$\begin{aligned} (ii) \text{ Vector projection of } \underline{b} \text{ onto } \underline{a} &= \frac{\underline{b} \cdot \underline{a}}{|\underline{a}|^2} \cdot \underline{a} \\ &= \frac{-4 + 2 + 12}{21} \cdot \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \\ &= \frac{10}{21} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \end{aligned}$$

(iii) If $\underline{b} = \lambda \underline{a} + \underline{c}$ then $\lambda \underline{a}$ is the vector projection of $\underline{b}$ onto $\underline{a}$

$$\therefore \lambda = \frac{10}{21}$$

$$\begin{aligned} \underline{c} &= \underline{b} - \lambda \underline{a} \\ &= \begin{bmatrix} -2 \\ -2 \\ 3 \end{bmatrix} - \frac{10}{21} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} -62/21 \\ -32/21 \\ 23/21 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} (d) \quad (i) \quad \overrightarrow{AB} &= \begin{bmatrix} -3+2 \\ 1+1 \\ 1-2 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \quad (1) \end{aligned}$$

$$\text{Vector equation} = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \quad (1)$$

$$\therefore x = -2 - \lambda \quad y = 2\lambda - 1 \quad z = 2 - \lambda \quad (1)$$

$$(ii) \quad \vec{OC} = \begin{bmatrix} 3 \\ 3 \\ -1 \end{bmatrix}$$

$$\text{When } x = 3, \quad \begin{aligned} 3 &= -2 - \lambda \\ \lambda &= -5 \end{aligned}$$

$$\text{When } \lambda = -5, \quad y = -11 \text{ \& } z = 7$$

$\therefore C$ does not lie on the line.

$$(a) \quad L_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, \quad L_2 = \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix} + \mu \begin{bmatrix} 3 \\ -4 \\ -2 \end{bmatrix}$$

$$(i) \quad \begin{array}{lcl} L_1 & & L_2 \\ x = 2 + 2\lambda & \longleftrightarrow & x = -2 + 3\mu \\ y = -1 + 2\lambda & \longleftrightarrow & y = 1 - 4\mu \\ z = 3 - \lambda & & z = -3 - 2\mu \end{array} \quad (1)$$

Solve x & y

$$\begin{array}{lcl} 2 + 2\lambda = -2 + 3\mu & & 2\lambda - 3\mu = -4 \quad \dots (1) \\ -1 + 2\lambda = 1 - 4\mu & & 2\lambda + 4\mu = 2 \quad \dots (2) \end{array}$$

$$(1) - (2)$$

$$-7\mu = -6$$

$$\mu = \frac{6}{7} \quad \dots (3)$$

$$(3) \Rightarrow (1)$$

$$2 \times \frac{6}{7} - 3\mu = -4$$

$$\frac{40}{7} = 3\mu$$

$$\mu = \frac{40}{21}$$

Check in z equations

$$L_1: z = 3 - \frac{6}{7}$$

$$L_2: z = -2 \times \frac{40}{21} - 3$$

$$z = \frac{15}{7}$$

$$= -\frac{23}{3}$$

Not the same. $\therefore$ No point of intersection.

$$(ii) \quad \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \neq k \begin{bmatrix} 3 \\ -4 \\ -2 \end{bmatrix} \quad k \in \mathbb{R} \quad \therefore \text{Not parallel}$$

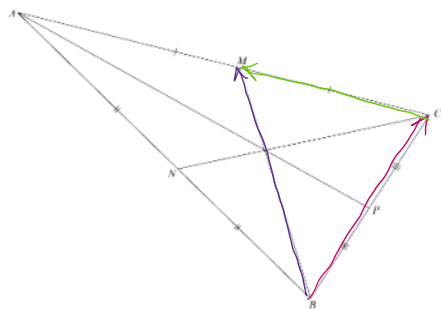
$$\begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -4 \\ -2 \end{bmatrix} = 6 - 8 + 2$$

$$= 0$$

$\therefore$ They are perpendicular, but do not intersect $\therefore$ skew & perpendicular.

(1)

(b)



(i) $\vec{OA} = \underline{a}$ $\vec{OB} = \underline{b}$ $\vec{OC} = \underline{c}$

$$\vec{CA} = \underline{a} - \underline{c} \quad \therefore \quad \vec{CM} = \frac{1}{2} (\underline{a} - \underline{c})$$

$$\vec{BC} = \underline{c} - \underline{b}$$

$$\therefore \vec{BM} = \vec{BC} + \vec{CM}$$

$$= \underline{c} - \underline{b} + \frac{1}{2} \underline{a} - \frac{1}{2} \underline{c}$$

$$= \frac{1}{2} \underline{a} + \frac{1}{2} \underline{c} - \underline{b}$$

$$= \frac{1}{2} (\underline{a} + \underline{c}) - \underline{b}$$

①

(ii) Generalising...

$$\vec{BM} = \frac{1}{2} (\underline{a} + \underline{c}) - \underline{b}$$

$$\vec{CN} = \frac{1}{2} (\underline{a} + \underline{b}) - \underline{c}$$

$$\vec{AP} = \frac{1}{2} (\underline{b} + \underline{c}) - \underline{a}$$

Generalise ①

$$\vec{BM} + \vec{CN} + \vec{AP} = \frac{1}{2} (\underline{a} + \underline{c}) - \underline{b} + \frac{1}{2} (\underline{a} + \underline{b}) - \underline{c} + \frac{1}{2} (\underline{b} + \underline{c}) - \underline{a}$$

$$= \left(\frac{1}{2} + \frac{1}{2} - 1 \right) \underline{a} + \left(\frac{1}{2} + \frac{1}{2} - 1 \right) \underline{b} + \left(\frac{1}{2} + \frac{1}{2} - 1 \right) \underline{c}$$

$$= 0 + 0 + 0 \quad \text{As required.}$$

① Show.

(c) $A(1, 4, -2) \quad B(-3, 0, 6) \quad C(-1, 2, -4)$

(i) let $\underline{n}$ be perpendicular to P_1

$$\therefore \underline{n} \cdot \vec{AB} = \underline{n} \cdot \vec{AC} = 0$$

$$\text{let } \underline{n} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \vec{AB} = \begin{bmatrix} -4 \\ -4 \end{bmatrix} \quad \vec{AC} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$\text{let } \underline{n} = \begin{bmatrix} 1 \\ y \\ z \end{bmatrix} \quad \overrightarrow{AB} = \begin{bmatrix} -4 \\ -4 \\ 8 \end{bmatrix} \quad \overrightarrow{AC} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$$

$$\underline{n} \cdot \overrightarrow{AB} = -4 - 4y + 8z = 0$$

$$\underline{n} \cdot \overrightarrow{AC} = -2 - 2y - 2z = 0$$

$$\therefore y + z = -1 \quad \dots \textcircled{1}$$

$$4y - 8z = -4 \quad \dots \textcircled{2}$$

$$\textcircled{2} - 4 \times \textcircled{1}$$

$$-12z = 0$$

$$z = 0$$

$$\therefore y = -1$$

$$\therefore \underline{n} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad \textcircled{1}$$

$$\text{Equation of } P_1: \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} x-1 \\ y-4 \\ z+2 \end{bmatrix} = 0$$

$$x-1 - (y-4) + 0 = 0$$

$$x-1-y+4 = 0$$

$$x-y = -3 \quad \textcircled{1}$$

(ii) If R, S & T are not sufficient then they must be collinear. $\textcircled{1}$

(iii) $\underline{n} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ let $\underline{m}$ be the vector perpendicular to P_3

$$\underline{m} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \quad \textcircled{1} \quad (\text{from equation})$$

Angle between $\underline{m}$ & $\underline{n}$ is the angle between the

Angle between $\underline{m}$ & $\underline{n}$ is the angle between the planes.

$$\cos \theta = \frac{\underline{m} \cdot \underline{n}}{|\underline{m}| |\underline{n}|}$$

$$|\underline{n}| = \sqrt{2}$$

$$|\underline{m}| = \sqrt{4}$$

$$\underline{m} \cdot \underline{n} = 3 \times 2 + 0$$

$$= 5$$

$$\therefore \cos \theta = \frac{5}{2\sqrt{2}}$$

$$\theta = 19^\circ$$

(1)

(d)

$$2x^2 + 2y^2 + 2z^2 - 4x + y - 12z = -2$$

$$x^2 - 2x + 1 + y^2 + \frac{y}{2} + \frac{1}{16} + z^2 - 6z + 9 = -1 + 1 + \frac{1}{16} + 9$$

$$(x-1)^2 + \left(y + \frac{1}{4}\right)^2 + (z-3)^2 = \frac{145}{16}$$

$$\text{centre: } \left(1, -\frac{1}{4}, 3\right), \text{ radius: } \frac{\sqrt{145}}{4} \approx 3.01$$

$$(a) (i) \underline{r} = \begin{bmatrix} 2\sin t \\ 2\cos t \\ \sin 2t \end{bmatrix}$$

$$\frac{x}{2} = \sin t$$

$$\frac{y}{2} = \cos t$$

$$z = 2\sin t \cos t$$

$$\therefore z = \frac{xy}{2} \quad (1)$$

$$(ii) \underline{r}'(t) = \begin{bmatrix} 2\cos t \\ -2\sin t \\ 2\cos 2t \end{bmatrix} \quad (1)$$

$$\text{@ } t = \frac{\pi}{6} \quad \underline{r}\left(\frac{\pi}{6}\right) = \begin{bmatrix} 1 \\ \sqrt{3} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 \\ 2\sqrt{3} \\ \sqrt{3} \end{bmatrix}$$

$$\underline{r}'\left(\frac{\pi}{6}\right) = \begin{bmatrix} \sqrt{3} \\ -1 \\ 1 \end{bmatrix} \quad (1)$$

$$\therefore \text{Equation of tangent} = \frac{1}{2} \begin{bmatrix} 2 \\ 2\sqrt{3} \\ \sqrt{3} \end{bmatrix} + \lambda \begin{bmatrix} \sqrt{3} \\ -1 \\ 1 \end{bmatrix} \quad \text{where } \lambda \in \mathbb{R}$$

$$\therefore x = 1 + \sqrt{3}\lambda \quad y = \sqrt{3} - \lambda \quad z = \frac{\sqrt{3} + \lambda}{2} \quad (1)$$

$$\frac{x-1}{\sqrt{3}} = \sqrt{3} - y = \frac{2z - \sqrt{3}}{2}$$

(iii) Gradient vector parallel to the x/y plane has

a 'z' component = 0

$$\therefore 2\cos 2t = 0$$

$$2t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \dots$$

$$t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$\therefore$ 4 points

$$P_{1/2} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \\ 0 \end{bmatrix}$$

$$t = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$P_{3/4} = \begin{bmatrix} \sqrt{2} \\ -\sqrt{2} \\ 0 \end{bmatrix}$$

$$t = \frac{3\pi}{4}, \frac{7\pi}{4} \quad (1)$$

$$L \quad - \quad \perp$$

$$t = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$L \quad - \quad -$$

$$t = \frac{3\pi}{4}, \frac{7\pi}{4}$$

(b) $r_1 = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$ $r_2 = \begin{bmatrix} 4 \\ 0 \\ -2 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}$

(i) $\therefore P = \begin{bmatrix} 3-p \\ 2p-4 \\ 3p+3 \end{bmatrix}$ $Q = \begin{bmatrix} 4+2q \\ -2q \\ -2-q \end{bmatrix}$

$$\therefore \vec{PQ} = \begin{bmatrix} 4+2q-(3-p) \\ -2q-(2p-4) \\ -2-q-(3p+3) \end{bmatrix}$$

$$= \begin{bmatrix} 1+2q+p \\ 4-2q-2p \\ -5-q-3p \end{bmatrix} \quad (1)$$

(ii) $\vec{PQ} \cdot \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} = 0$ $\&$ $\vec{PQ} \cdot \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} = 0$ (1)

(1) (2)

(1) $\therefore -(1+2q+p) + 2(4-2q-2p) + 3(-5-q-3p) = 0$

$$-8 - 9q - 14p = 0$$

$$14p + 9q = -8 \quad \dots (1)$$

(2) $2(1+2q+p) - 2(4-2q-2p) - (-5-q-3p) = 0$ (1)

$$-1 + 9q + 9p = 0$$

$$9p + 9q = 1 \quad \dots (2)$$

(1) - (2)

$$5p = -9$$

$$p = -9/5$$

$$\therefore q = 86/45 \quad] (1)$$

(iii) Minimum distance occurs when $\vec{PQ}$ is perpendicular to both lines.

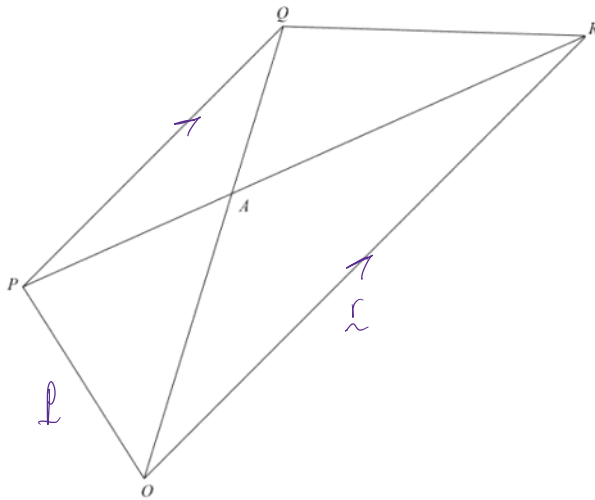
$$\therefore \vec{PQ} = \begin{bmatrix} 136/45 \\ 34/9 \\ -68/45 \end{bmatrix}$$

Minimum distance = $|\vec{PQ}|$

$$\therefore \text{Minimum distance} = |\vec{PQ}|$$

$$\therefore 9.3535 \dots$$

(c)



$$(1) \vec{OQ} = \vec{OP} + \vec{PQ}$$

$\vec{PQ}$ is parallel to $\vec{OR}$

$$\therefore \vec{OQ} = \vec{OP} + \lambda \vec{OR} \quad \lambda \in \mathbb{R}$$

$$= \vec{p} + \lambda \vec{r}$$

The line OQ is parallel to $\vec{OQ}$ but goes through origin

$$\therefore \vec{r}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t(\vec{OQ})$$

$$= t(\vec{p} + \lambda \vec{r}) \quad \text{As required.}$$

(ii) let the midpoint of $\vec{PQ}$ be M
 $\vec{OR}$ be N

$\therefore$ R.P A lies on $\vec{MN}$

Equation of the line PR = l_2

$$l_2 = \vec{OP} + k \vec{PR}$$

$$= \vec{p} + k(\vec{r} - \vec{p})$$

A is the point of intersection between those lines.

∴ Solve simultaneously.

$$p + k(r - p) = t(p + \lambda r)$$

$$p(1 - k) + kr = tp + \lambda tr$$

As p & r are not parallel

$$1 - k = t \dots \textcircled{1} \quad \& \quad k = \lambda t \dots \textcircled{2}$$

$$\textcircled{2} \Rightarrow \textcircled{1}$$

$$1 - \lambda t = t$$

$$1 = \lambda t + t$$

$$t = \frac{1}{1 + \lambda} \quad \therefore \quad k = \frac{\lambda}{1 + \lambda}$$

$$\therefore \vec{OP} = t(p + \lambda r) \quad \text{As } p \text{ lies on } l_1$$

$$= \frac{1}{1 + \lambda} (p + \lambda r)$$

$$= \frac{1}{1 + \lambda} p + \frac{\lambda}{1 + \lambda} r \quad \textcircled{1}$$

$$\vec{OM} = \vec{OP} + \frac{1}{2} \vec{PQ} \quad \vec{ON} = \frac{1}{2} r$$

$$= p + \frac{1}{2} \lambda r$$

$$\vec{NM} = p + \frac{1}{2} \lambda r - \frac{1}{2} r$$

$$= p + \frac{1}{2} (\lambda - 1) r$$

$$\text{suppose } \vec{NP} = \mu \vec{NM}$$

$$\begin{aligned} \therefore \vec{OP} &= \vec{ON} + \vec{NM} \\ &= \frac{1}{2} r + \mu \left[p + \frac{1}{2} (\lambda - 1) r \right] \\ &= \mu p + \left[\frac{1}{2} + \frac{\mu}{2} (\lambda - 1) \right] r \quad \textcircled{1} \end{aligned}$$

$$\text{If } \mu = \frac{1}{1 + \lambda}$$

$$\begin{aligned} \text{Then } \vec{OP} &= \frac{1}{1 + \lambda} p + \left[\frac{1}{2} + \frac{1}{2} \left(\frac{1}{1 + \lambda} \right) (\lambda - 1) \right] r \\ &= \frac{1}{1 + \lambda} p + \left[\frac{(1 + \lambda) + (\lambda - 1)}{2(1 + \lambda)} \right] r \end{aligned}$$

$$= \frac{1}{1+\lambda} \vec{p} + \left[\frac{(1+\lambda) + (\lambda-1)}{2(1+\lambda)} \right] \vec{q}$$

$$= \frac{1}{1+\lambda} \vec{p} + \frac{2\lambda}{2(1+\lambda)} \vec{q}$$

$$= \frac{1}{1+\lambda} \vec{p} + \frac{\lambda}{(1+\lambda)} \vec{q} = \vec{OP} \text{ from point of intersection}$$

①